FDVA Mini Project

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Sample Superstore Mini Project

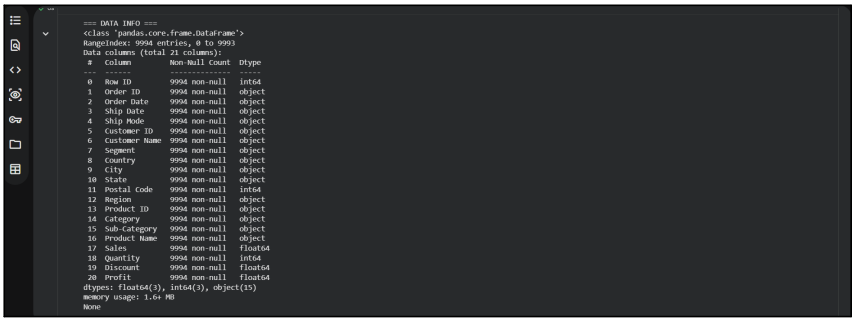
Aim:

To analyse the Sample Superstore dataset using Exploratory Data Analysis (EDA) in Python, and to visualise sales and profit trends using Python plots, Power BI and Tableau.

Procedure:

1. Exploratory Data Analysis (EDA)

Output 1: Dataset information

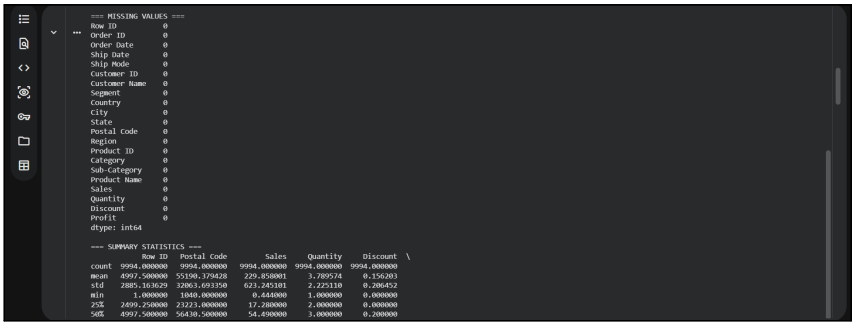


```
=== DATA INFO ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
None
```

Used: pandas `df.info()` to view the structure of the data.

Result: The dataset has 9,994 rows and 21 columns with no null values in any column. It contains 15 text (object), 3 integer and 3 float columns. Order Date and Ship Date are stored as text, so they need conversion to date type.

Output 2: Missing values and summary statistics



```
=== MISSING VALUES ===
Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code      0
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
Quantity         0
Discount         0
Profit           0
dtype: int64

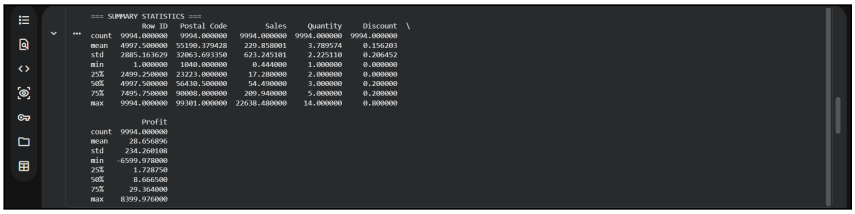
=== SUMMARY STATISTICS ===
             Row ID   Postal Code         Sales     Quantity     Discount  \
count   9994.000000   9994.000000   9994.000000  9994.000000  9994.000000
mean    4997.500000  55190.379428    229.858001     3.789574     0.156203
std     2885.163629  32063.693350    623.245101     2.225110     0.206452
min        1.000000   1040.000000      0.444000     1.000000     0.000000
25%     2499.250000  23223.000000     17.280000     2.000000     0.000000
50%     4997.500000  56430.500000     54.490000     3.000000     0.200000

             Profit
count   9994.000000
mean      28.656896
std      234.260108
min    -6599.978000
25%        1.728750
50%        8.666500
75%       29.364000
max     8399.976000
```

Used: `df.isnull().sum()` and `df.describe()`.

Result: All columns show 0 missing values, so the data is clean and needs no imputation. The average sale is about 229.86 per order line, with an average quantity of 3.79 and an average discount of about 15.6%.

Output 3: Summary statistics (continued)



```
=== SUMMARY STATISTICS ===
             Row ID   Postal Code         Sales     Quantity     Discount  \
count   9994.000000   9994.000000   9994.000000  9994.000000  9994.000000
mean    4997.500000  55190.379428    229.858001     3.789574     0.156203
std     2885.163629  32063.693350    623.245101     2.225110     0.206452
min        1.000000   1040.000000      0.444000     1.000000     0.000000
25%     2499.250000  23223.000000     17.280000     2.000000     0.000000
50%     4997.500000  56430.500000     54.490000     3.000000     0.200000
75%     7495.750000  90008.000000    209.940000     5.000000     0.200000
max     9994.000000  99301.000000  22638.480000    14.000000     0.800000

             Profit
count   9994.000000
mean      28.656896
std      234.260108
min    -6599.978000
25%        1.728750
50%        8.666500
75%       29.364000
max     8399.976000
```

Used: `df.describe()` (min, quartiles, max for each numeric column).

Result: Sales range from 0.44 to 22,638.48, and the mean (229.86) is far above the median (54.49), so sales are highly skewed. Profit ranges from -6,599.98 to 8,399.98, which shows some orders make heavy losses.

2. Data Visualization (Python)

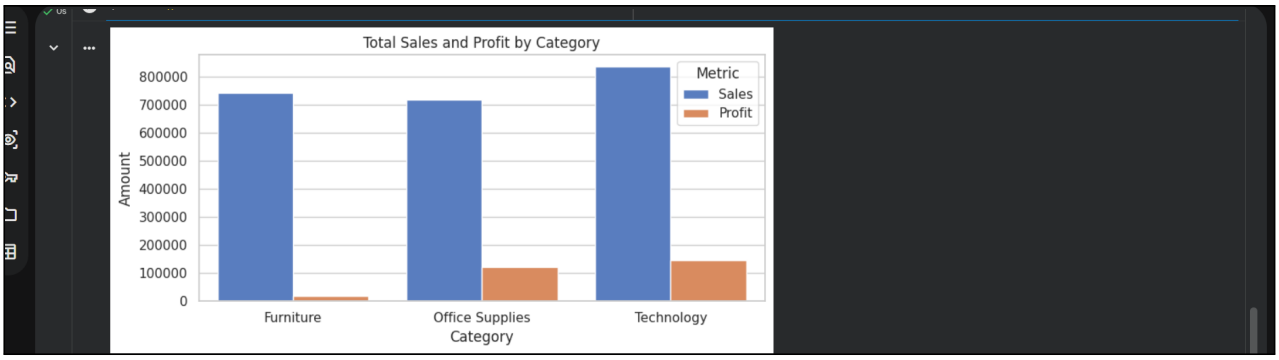
Plot 1: Sales Distribution



Used: Histogram with a KDE curve (seaborn `histplot`).

Result: Most orders have very small sales values, close to 0 to 500. The long tail towards the right (up to 20,000+) shows a few very large orders, so the sales distribution is right-skewed.

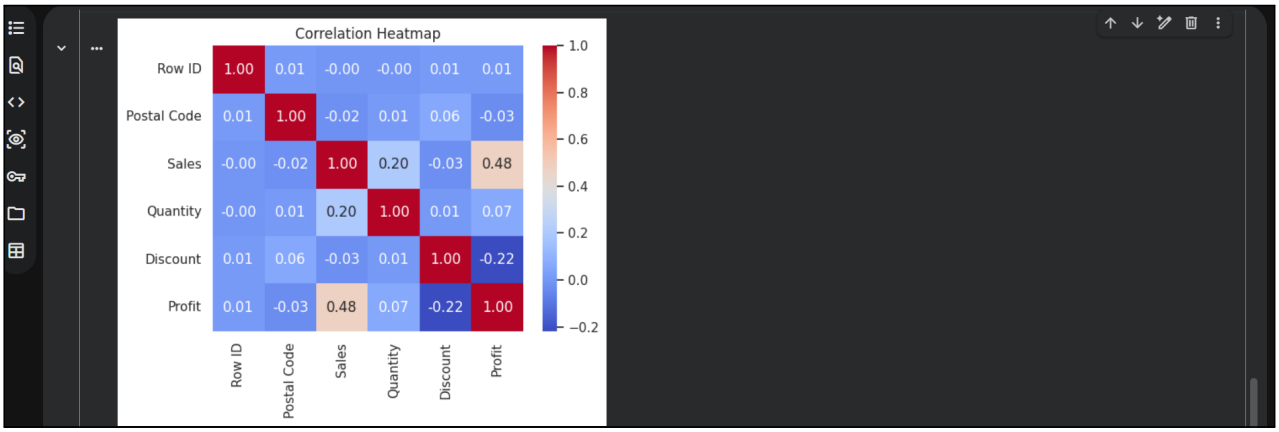
Plot 2: Total Sales and Profit by Category



Used: Grouped (clustered) bar chart comparing Sales and Profit for each Category (seaborn `barplot`).

Result: Technology has the highest sales and profit. Furniture has high sales (about 742K) but a very low profit, while Office Supplies gives a better profit from similar sales.

Plot 3: Correlation Heatmap

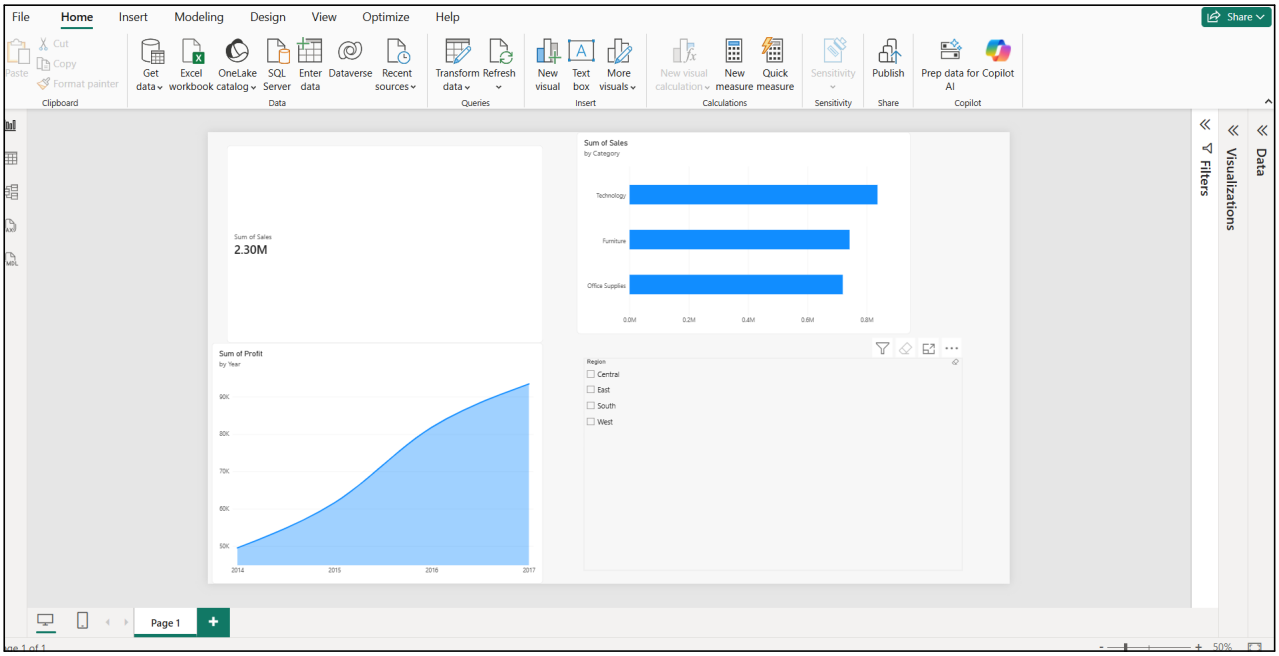


Used: Correlation heatmap with the coolwarm colour map (seaborn `heatmap`).

Result: Sales and Profit have a moderate positive correlation (0.48), and Sales and Quantity a weak one (0.20). Discount and Profit have a negative correlation (-0.22), so higher discounts tend to reduce profit.

3. Power BI

Power BI Output: Sales and Profit dashboard

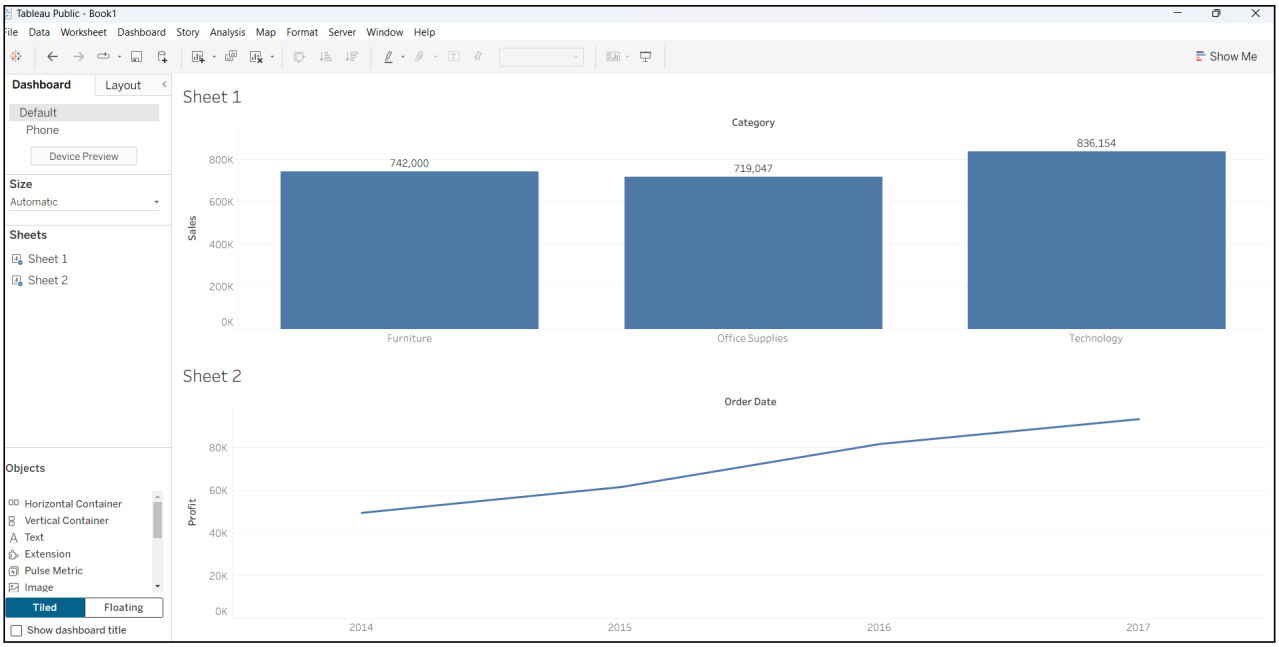


Used: Card (total sales), clustered bar chart (Sum of Sales by Category), area chart (Sum of Profit by Year) and a slicer (Region).

Result: Total sales are 2.30M, with Technology as the top category, followed by Furniture and Office Supplies. Profit grows steadily from about 50K in 2014 to over 90K in 2017, and the Region slicer filters the whole dashboard.

4. Tableau

Tableau Output: Sales by Category and Profit by Year dashboard



Used: Bar chart (Sales by Category, Sheet 1) and line chart (Profit by Order Date year, Sheet 2) combined in Dashboard 1.

Result: Technology leads with 836,154 in sales, then Furniture (742,000) and Office Supplies (719,047). The profit line shows a continuous rise from 2014 to 2017, which shows the business is growing each year.